

## I. Document Change log

| <b>Date</b>       | <b>Version</b> | <b>Description</b>                      | <b>Author</b>        |
|-------------------|----------------|-----------------------------------------|----------------------|
| <b>12/30/2025</b> | <b>1.0</b>     | Created document                        | Diep Tu Huy          |
| <b>1/1/2026</b>   | <b>1.1</b>     | Add Function/Non-Functional Requirement | Tran Thao Nhi        |
| <b>1/1/2026</b>   | <b>1.2</b>     | Add DFD Level 0                         | Diep Tu Huy          |
| <b>1/2/2026</b>   | <b>1.3</b>     | Add DFD Level 1                         | Diep Tu huy          |
| <b>1/3/2026</b>   | <b>1.4</b>     | Add Usecase Diagram                     | Nguyen Thi Quynh Nhu |
| <b>1/3/2026</b>   | <b>1.5</b>     | Add Usecase Description                 | Nguyen Thanh Huy     |
| <b>1/4/2026</b>   | <b>1.6</b>     | Add Class Diagram                       | Tran Tan Phat        |
| <b>1/4/2026</b>   | <b>1.7</b>     | Add Data Model                          | Tran Thao Nhi        |
| <b>1/5/2026</b>   | <b>1.8</b>     | Add Data Model                          | Tran Tan Phat        |
| <b>1/5/2026</b>   | <b>1.9</b>     | Add DFD Level 2                         | Diep Tu Huy          |
| <b>1/6/2026</b>   | <b>1.10</b>    | Add UI Design                           | Nguyen Thanh Huy     |

## Library Management System (LMS)

### 1. Functional Requirements

#### 1.1 User Management

##### Login/Logout

- Users log in using their username and password
- The system supports secure logout
- The system displays an error message if the login information is incorrect

## **Account Management (Administrator)**

- Create new accounts for different roles
- Update user account information
- Suspend or reactivate user accounts
- Delete user accounts from the system

## **Access Control**

- The system automatically classifies and restricts functions based on roles
- Administrator: Full system management privileges
- Librarian: Manage books, process borrow/return transactions, manage fines
- Member: Can only search for books, borrow/return books, view personal history

## **1.2 Book Management**

### **Add New Books (Librarian/Administrator)**

- Enter complete book information: Book ID, Title, Author, Publisher, Publication Year, Category, ISBN, Quantity
- The system checks for duplicate ISBN or Book ID

### **Update Book Information (Librarian/Administrator)**

- Edit information for existing books in the system
- Update book quantity when new copies are added

### **Delete Books (Librarian/Administrator)**

- Remove books from the catalog when necessary
- The system verifies if the book is currently borrowed before deletion

### **Search Books (All users)**

- Search by: Title, Author, Category, ISBN
- Display results with full information and available quantity

## **1.3 Borrowing and Returning Books**

### **Borrow Books (Member)**

- Members select available books to borrow
- The system automatically records: Borrow Date, Due Date (typically 14 days later)
- Automatically decreases the available quantity in inventory
- Borrowing Restrictions

### **Prevent borrowing when no copies are available**

- Limit the number of books borrowed simultaneously (e.g., maximum of 5 books)
- Check if the member has unpaid fines before allowing new borrowings

### **Return Books (Member/Librarian)**

- Record the actual return date
- Automatically increase the available quantity in inventory
- Automatically calculate fines if returned past the due date

## **1.4 Fine Management**

### **Automatic Calculation**

- The system automatically calculates fines based on the number of overdue days
- Stores fine information along with the borrowing transaction

### **Track and Process Fines (Librarian)**

- View a list of unpaid fines
- Mark fines as paid when the member settles the payment
- Generate statistics on total fines over a period

## **1.5 Reports and Statistics**

### **Reports for Librarian/Administrator**

- List of currently borrowed books
- List of overdue books not yet returned
- List of active members
- Borrowing statistics by book/category
- Fine revenue statistics

### **System Statistics (Administrator)**

- Number of users by role
- Total number of books in inventory
- Percentage of books currently on loan
- System activity over time

## **2. Non-Functional Requirements**

### **2.1 Performance**

- The system responds to user requests within 3 seconds under normal load
- Supports at least 100 concurrent users
- Page load and transaction processing time under 2 seconds

### **2.2 Security**

- Password Encryption: User passwords are encrypted before storage
- Access Control: Detailed permission management based on user roles
- Data Protection: Prevent unauthorized access to the database
- Session Security: Automatic logout after a period of inactivity

### **2.3 Usability**

- Intuitive, user-friendly interface for all user types
- Clear menu structure with logical grouping of functions
- Specific, easy-to-understand error messages and guidance
- Support for quick searches with smart filters

### **2.4 Reliability**

- Ensures data consistency in all transactions

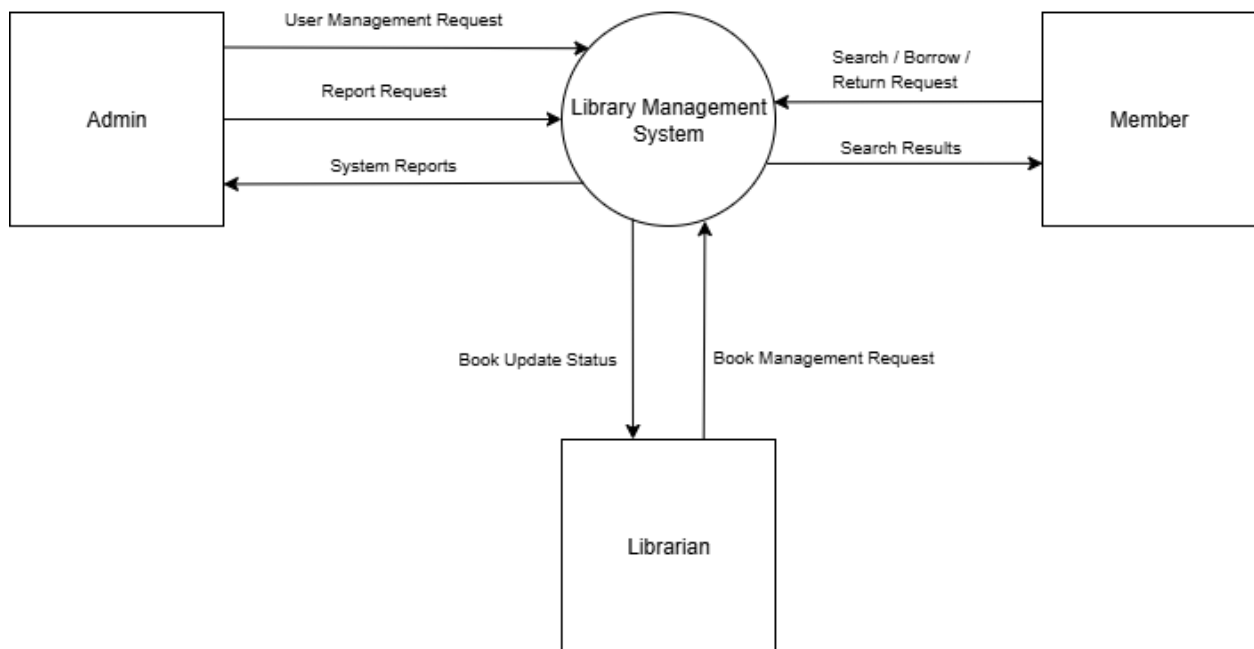
- Graceful error handling without data loss
- System operates stably 24/7 with minimal downtime
- Automatic periodic data backup

## 2.5 Maintainability

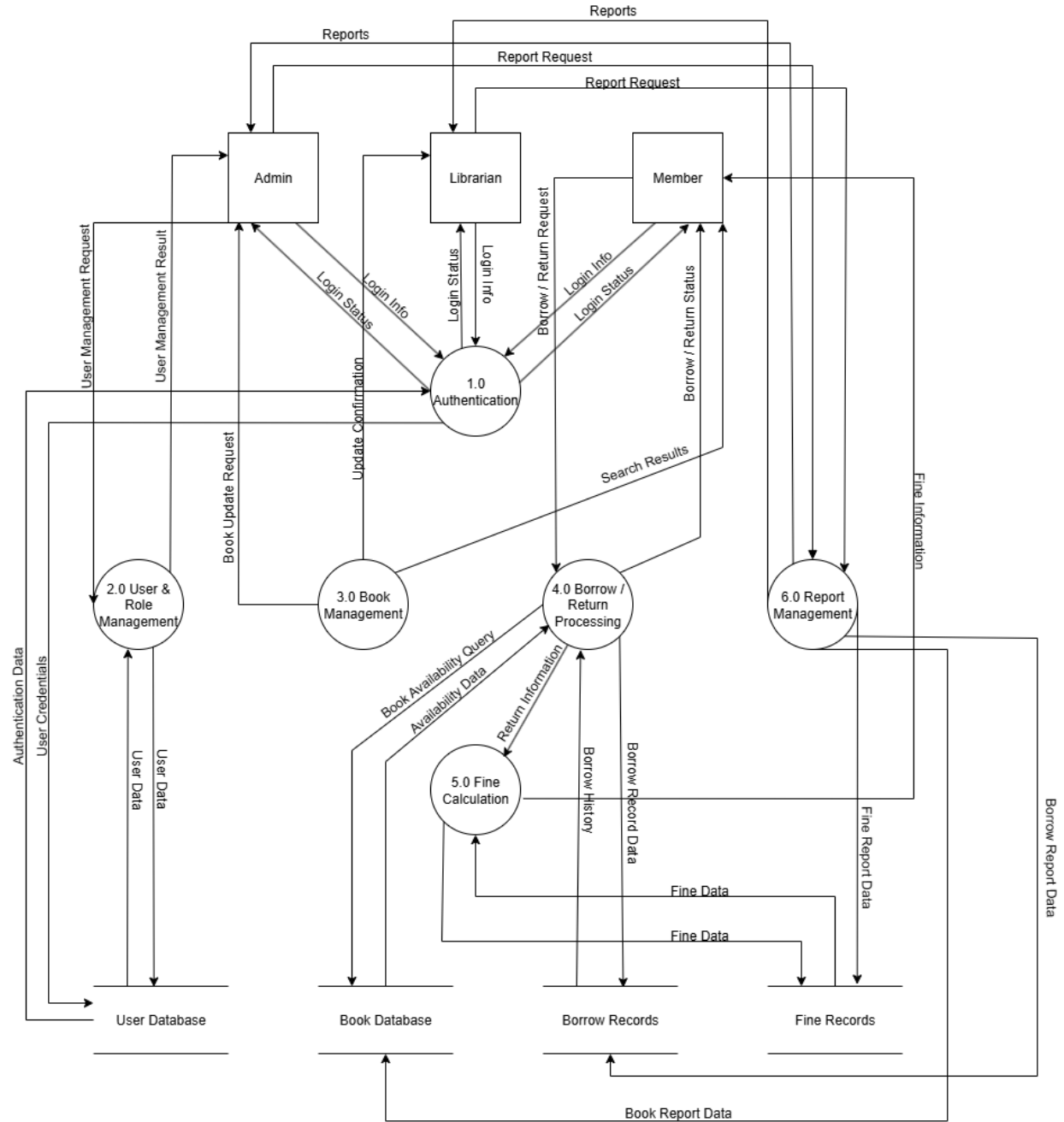
- Modular Architecture: Easy to add/remove features without affecting the system
- Clean Source Code: Fully documented, with meaningful variable and function names
- Easy Debugging: Detailed logging mechanism for system activities
- Scalability: Designed to allow scaling of user numbers and data volume

## 3. Data Flow Diagram (DFD)

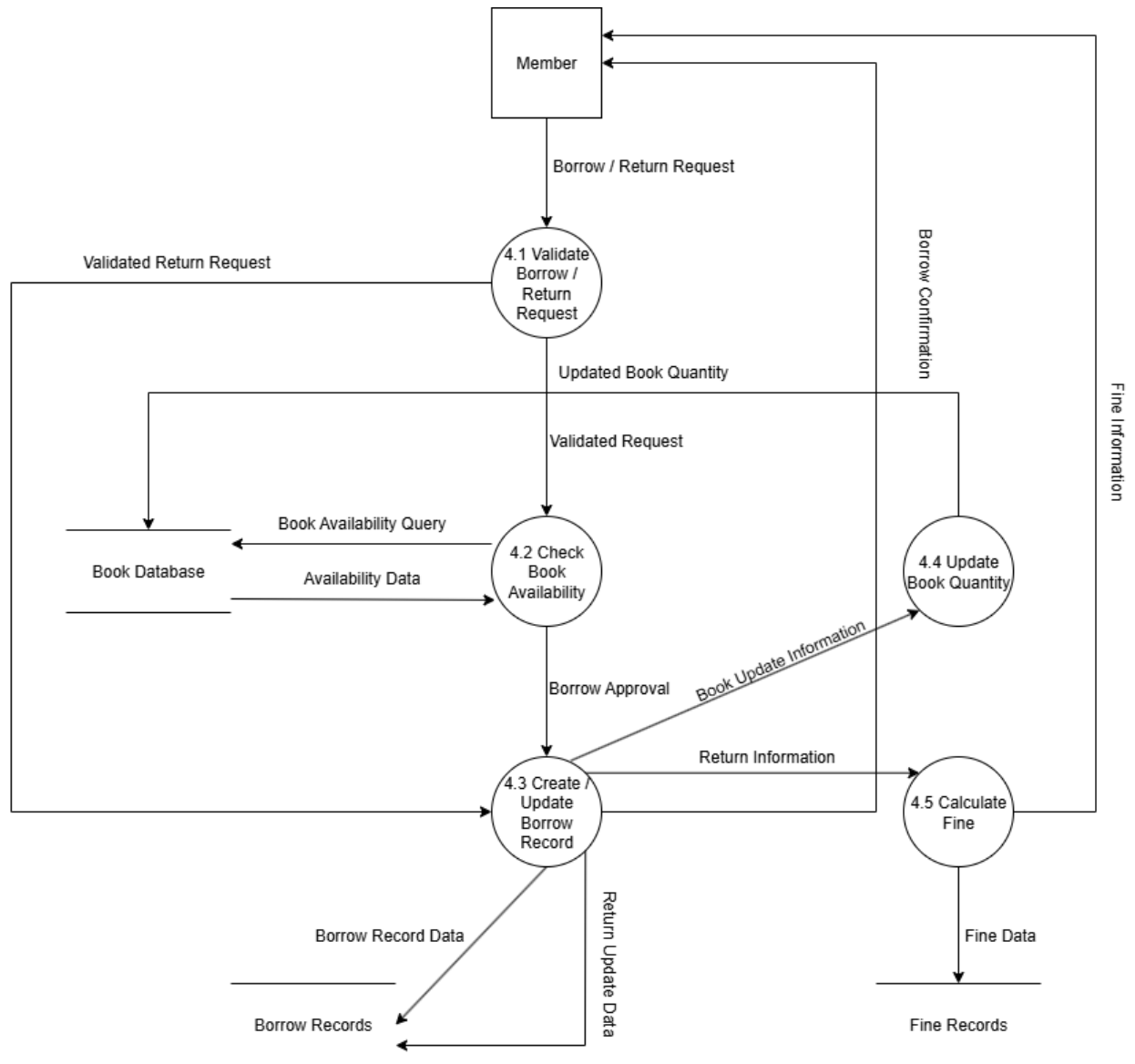
### 3.1 Level 0



### 3.2 Level 1 DFD

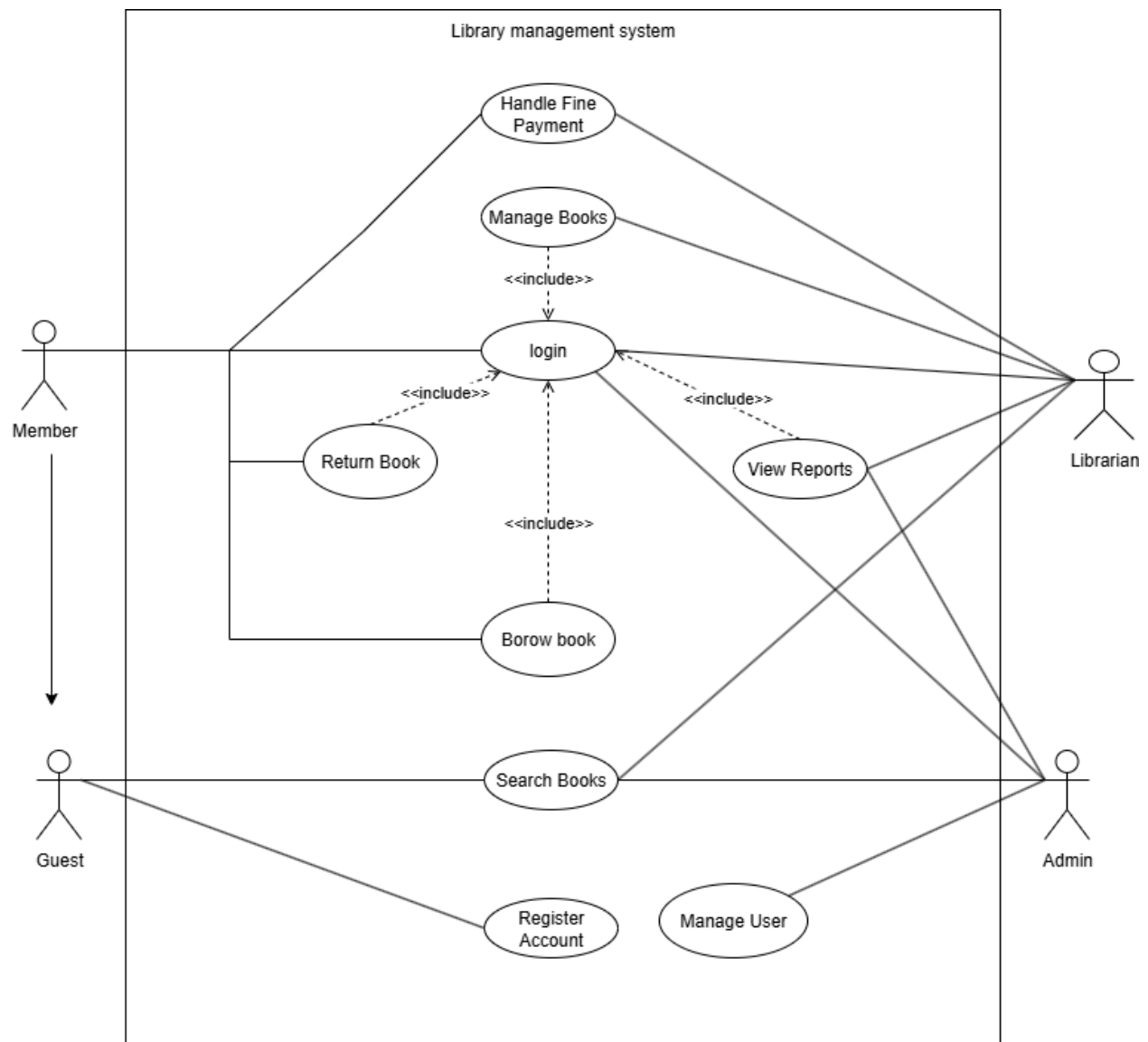


### 3.3 Level 2 DFD



## 4. Use Case Diagram

### 4.1 Diagram overview



#### List of Actors

| No. | Actor     | Meaning                                                                                          |
|-----|-----------|--------------------------------------------------------------------------------------------------|
| 1   | Guest     | Unregistered user who can only search for books and register for an account.                     |
| 2   | Member    | Registered user (Student/Faculty) who can borrow books, return books, and view personal history. |
| 3   | Librarian | Library staff responsible for managing books, processing transactions, and handling fines.       |
| 4   | Admin     | System administrator with full privileges to manage user accounts and system settings.           |

## List of Use Cases

| No. | Uses Case main      | Note (Group)    |
|-----|---------------------|-----------------|
| 1   | Login               | Authentication  |
| 2   | Register Account    | Authentication  |
| 3   | Search Books        | Book Management |
| 4   | Manage Books        | Book Management |
| 5   | Borrow Books        | Circulation     |
| 6   | Return Books        | Circulation     |
| 7   | Handle Fine Payment | Fine Management |
| 8   | View Reports        | Reporting       |
| 9   | Manage User         | User Management |

### 4.2 Use Case: Login

#### Summary

This use case allows all users (Member, Librarian, Administrator) to access the system. It authenticates credentials and checks for account lock status before granting access.

#### Flow of Events

##### Main Flow

- The use case starts when the user accesses the Login screen.
- The user enters their Username and Password.
- The system validates the credentials against the database.
- Include: The system performs a Lock User Account check to ensure the account is not currently suspended.
- If valid and active, the system grants access and redirects the user to their respective dashboard.

#### Alternate Flows

Invalid Credentials: The system displays an error message "Incorrect username or password" and asks the user to retry.

Account Locked: If the account status is "Locked", the system denies access.

#### Special Requirements

Passwords must be encrypted before storage and comparison.

#### Preconditions

The user is not logged in.

#### Postconditions

The user session is established.

### **Extension Points**

Change Password: Users may choose to change their password upon logging in.

## **4.3 Use Case: Register Account**

### **Summary**

This use case allows a Guest to register for a new Member account to start using library services.

### **Flow of Events**

#### Main Flow

- The Guest selects the "Register" option on the landing page.
- The system displays the registration form.
- The Guest enters the required personal information (Name, Email, etc.).
- The system validates the input format.
- Extend: The system triggers Create User Account to save the new record in the database.
- The system displays a success message.

### **Alternate Flows**

Existing Account: If the email already exists, the system notifies the Guest.

### **Special Requirements**

None.

### **Preconditions**

The user is a Guest (not logged in).

### **Postconditions**

A new Member account is created in the database.

## **4.4 Use Case: Search Books**

### **Summary**

This use case allows all users (Guest, Member, Librarian, Admin) to search for books in the library catalog using various criteria.

### **Flow of Events**

#### Main Flow

- The user selects the "Search Books" function.
- The user enters a keyword.
- Extend: The user may choose specific filter criteria:

- + Search by Title
- + Search by Author
- + Search by Category
- + Search by ISBN
- The system queries the database.
- The system displays a list of matching books.

### **Special Requirements**

None.

### **Preconditions**

The system database is accessible.

### **Postconditions**

Search results are displayed to the user.

## **4.5 Use Case: Manage Books**

### **Summary**

This use case allows Librarians or Admins to maintain the book catalog, including adding new books or updating existing ones.

### **Flow of Events**

Main Flow (Add New Book)

- The Librarian selects "Add New Book".
- The Librarian enters book details (Title, ISBN, Quantity, etc.).
- Include: The system performs Check Duplicate ISBN to ensure uniqueness.
- Include: The system performs Validate Book Information.
- The system saves the new book to the database.

### **Special Requirements**

None

### **Alternate Flows**

Edit/Remove (Extension Points): The Librarian can select a book to Update Book Information, Update Book Quantity, or Remove Book.

Remove Book Constraint: If "Remove Book" is selected but the book is currently borrowed, the system prevents.

User is logged in as Librarian or Admin.

**Preconditions**

User is logged in as Librarian or Admin.

**Postconditions**

The book database is updated.

**4.6 Use Case: Borrow Books****Summary**

This use case describes the process of checking out a book. It enforces business rules regarding inventory and borrowing limits.

**Flow of Events****Main Flow**

- The Member (or Librarian) initiates the borrowing process for a book.
- Include: The system executes Check Book Availability to verify the book is in stock.
- Include: The system executes Check Member Borrow Limit to ensure the member has not exceeded their quota.
- The system creates a Borrow Record with the current date.
- The system updates the book status and decrements the inventory count.
- The system confirms the transaction.

**Alternate Flows**

Unavailable/Limit Reached: If the book is out of stock or the member limit is reached, the system denies the request.

**Special Requirements**

Transaction processing must ensure data consistency.

**Preconditions**

Member is logged in and eligible to borrow.

**Postconditions**

Inventory is updated; Borrow Record is created.

**4.7 Use Case: Return Books****Summary**

This use case handles the return of books and automatically checks for overdue status to calculate fines if necessary.

## **Flow of Events**

### **Main Flow**

- The user scans the book to return.
- The system retrieves the active Borrow Record.
- The system updates the Return Date.
- Include: The system performs a Check Overdue verification.
- If not overdue, the system marks the book as "Available" and increments the inventory.

### **Alternate Flows**

Overdue Return (Extension Point): If the book is returned late (Check Overdue is true), the use case extends to Calculate Fine. The system calculates the penalty and records it.

### **Special Requirements**

None

### **Preconditions**

The book was previously borrowed.

### **Postconditions**

Book is returned to inventory. Fine is recorded if applicable.

## **4.8 Use Case: Handle Fine Payment**

### **Summary**

This use case allows the Librarian to process fine payments for Members with outstanding debts.

### **Flow of Events**

#### **Main Flow**

- The Librarian accesses the View Unpaid Fines function for a Member.
- The system displays the total outstanding amount.
- The Librarian selects Confirm Fine Payment after receiving the payment.
- Include: The system triggers Make as Paid.
- The system updates the fine status to "Paid" in the database.

### **Special Requirement**

None

### **Preconditions**

Member has unpaid fines.

**Postconditions**

Fine status is updated to Paid.

**4.9 Use Case: View Reports****Summary**

This use case allows Administrators and Librarians to generate and view various system reports.

**Flow of Events**

Main Flow

- The user selects the "Generate Reports" dashboard.
- The user chooses a specific report type.
- Include: The system can generate specific reports such as:
  - + View User List
  - + Report on Locked User Accounts
  - + Report on User Role
- Alternatively, the Admin can view View System Usage Statistics.
- The system retrieves data and displays the report.

**Special Requirements**

None

**Preconditions**

User is logged in with appropriate privileges (Admin or Librarian).

**Postconditions**

The requested report is displayed.

**4.10 Use Case: Manage User****Summary**

This use case allows the Administrator to manage user accounts within the system.

**Flow of Events**

Main Flow

- The Administrator selects the "Manage User Accounts" function.
- The Administrator selects a specific user account.
- The Administrator performs an action. This use case extends to:
  - + Create User Account: Add a new Librarian or Admin.
  - + Lock User Account: Suspend a user.

- + Unlock User Account: Restore access.
- + Delete User Account: Remove a user.
- + Assign User Role: Change user permissions.
- The system executes the action and updates the database.

**Special Requirements**

None

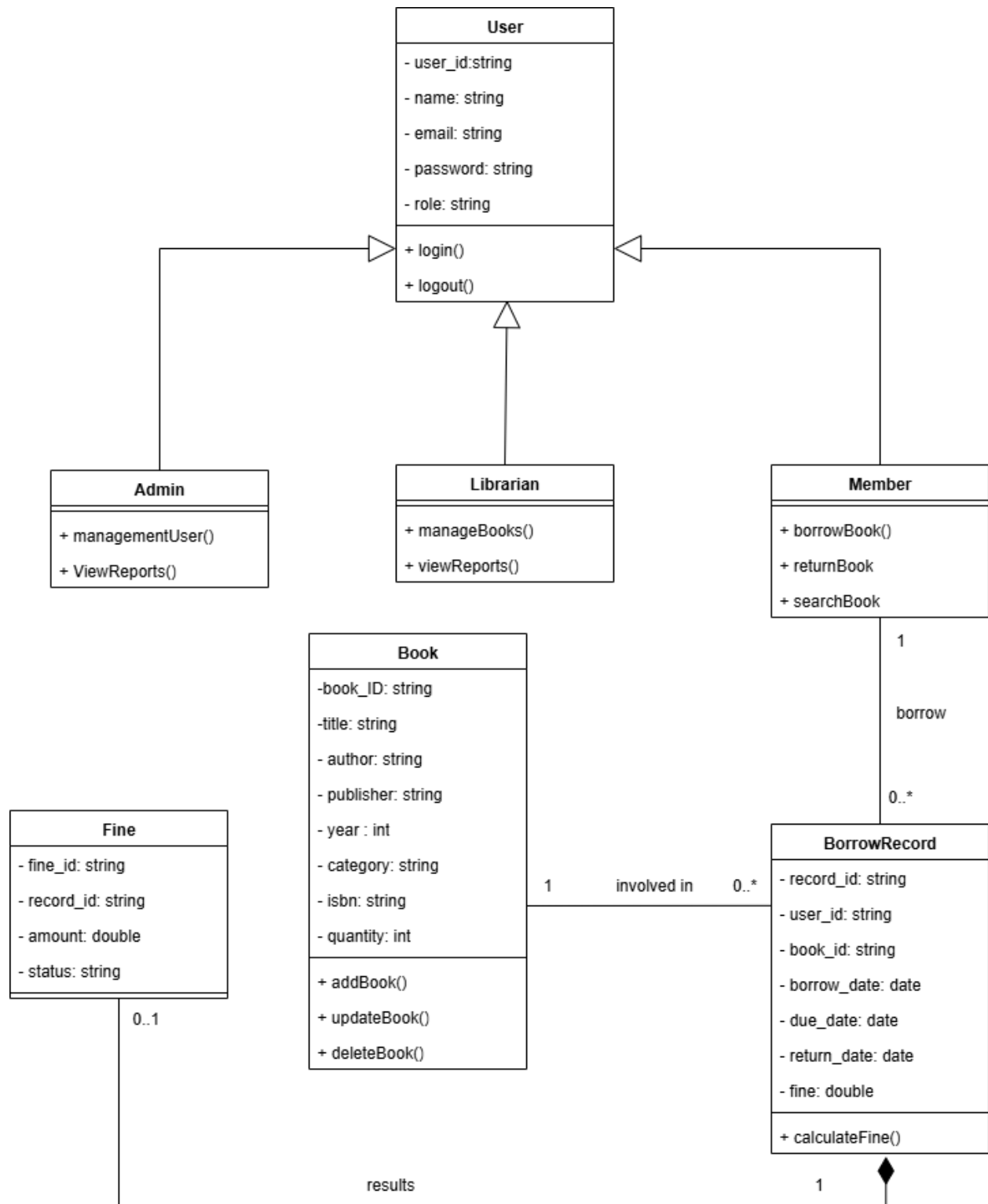
**Preconditions**

User is logged in as Administrator.

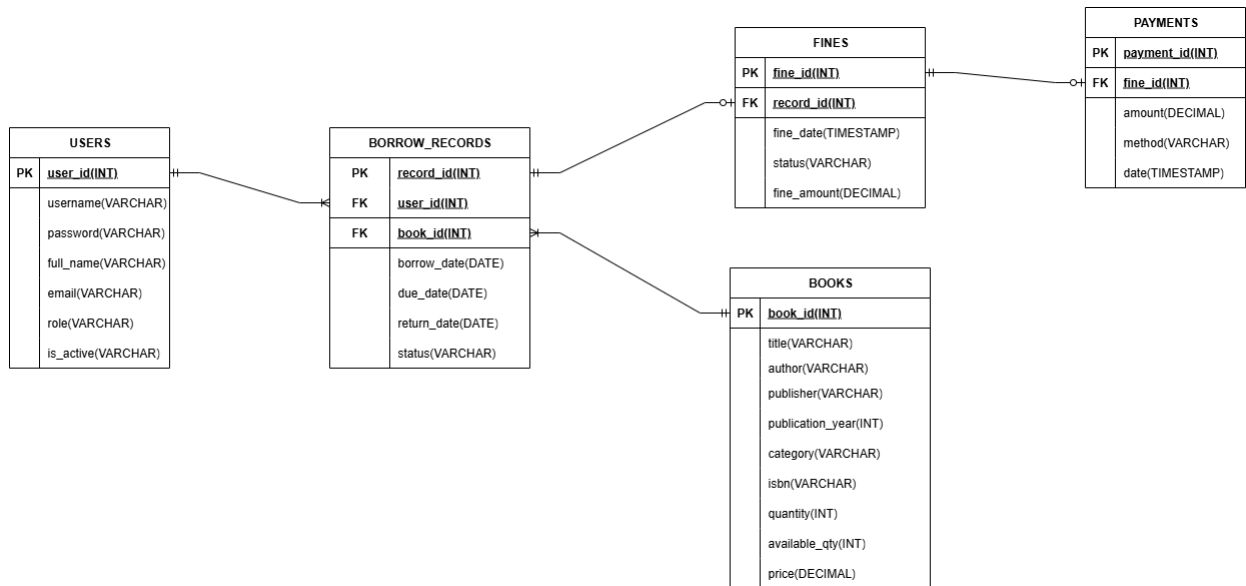
**Postconditions**

User account status is modified.

## 5. Class Diagram



## 6. Data Model – Diagram Overview



## 7. Interface Design Description

### 7.1 Login Interface

#### Description

The Login Interface allows users to authenticate and access the system based on their assigned roles.

#### Main Components

- Username input field
- Password input field
- Login button
- Error message area (displayed when login fails)

#### Behavior

- Users must enter a valid username and password to log in.
- The system validates the credentials against the User Database.
- Upon successful login, users are redirected to the Dashboard Interface corresponding to their role.
- If authentication fails, an error message is displayed.

### 7.2 Dashboard Interface

#### Description

The Dashboard Interface serves as the main navigation screen after login. The displayed features depend on the user role.

## Main Components

- Navigation menu (role-based)
- Summary information (e.g., number of borrowed books, overdue books)
- Quick access buttons to main functionalities

## Role-Based Access

- **Administrator:**
  - User Management
  - Book Management
  - Reports and Statistics
- **Librarian:**
  - Book Management
  - Fine Management
  - Reports
- **Member:**
  - Search Books
  - Borrow Books
  - Return Books
  - View Borrowing History
  - View Fines

## 7.3 Book Management Interface

### Description

The Book Management Interface allows administrators and librarians to manage book information in the library.

### Main Components

- Book list table displaying:
  - Book ID, Title, Author, Category, Quantity
- Search bar (search by title, author, category, ISBN)
- Add Book button
- Edit Book button
- Delete Book button

### Behavior

- Authorized users can add new books by entering required book information.
- Existing book records can be updated or removed.
- Search results are displayed dynamically based on user input.

## **7.4 Borrow / Return Interface**

### **Description**

The Borrow / Return Interface enables members to borrow available books and return previously borrowed books.

### **Main Components**

- List of available books
- Borrow button
- List of borrowed books
- Return button
- Borrowing details (borrow date, due date, return date)

### **Behavior**

- The system checks book availability and borrowing limits before allowing borrowing.
- Upon borrowing, the system records the transaction and updates book quantity.
- When returning a book, the system updates the return date and triggers fine calculation if overdue.

## **7.5 Fine Management Interface**

### **Description**

The Fine Management Interface allows librarians to manage fines and members to view their fine information.

### **Main Components**

- Fine list table (borrow record, fine amount, status)
- Fine status indicator (Paid / Unpaid)

### **Behavior**

- The system automatically calculates fines for overdue returns.
- Librarians can view and manage unpaid fines.
- Members can view their own fine details.

**Related Use Cases:** Calculate Fine, View Fine Information

**Related DFD Process:** Fine Calculation Processing (Process 5.0)

## **7.6 Report Interface**

### **Description**

The Report Interface provides reporting and statistical information for administrators and librarians.

## **Main Components**

- Report selection options:
  - Borrowed Books Report
  - Overdue Books Report
  - Active Members Report
- Display area for generated reports

## **Behavior**

- Users can select a report type to view corresponding data.
- Reports are generated based on data stored in the system.
- Administrators can view overall system usage statistics.

**Related Use Cases:** View Reports

**Related DFD Process:** Report Management (Process 6.0)